MONOTONICITY OF ERDŐS SUMS

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ABSTRACT. For $k \geq 1$ let $f_k = \sum_{\Omega(n)=k} 1/(n \log n)$, where Ω counts prime factors with multiplicity. We prove that $f_k < f_{k+1}$ for every $k \geq 6$, whereas the sums g_k of $1/(n \log n)$ over odd n with $\Omega(n) = k$ satisfy $g_k > g_{k+1}$ for every $k \geq 1$. The two statements arise from the same analytic distinction. Writing the generating function $\sum_k P_k(s) z^k$ of the associated Dirichlet series as $\zeta(s)^z H(s, z)$, at $s = 1$ the point $z = 2$ lies on the first singular circle for the unrestricted product, and strictly inside the first singular circle of the product with the prime 2 removed. The analytic reductions are written explicitly below. The finite inequalities and coefficient bounds used in the last step are certified by ball arithmetic, and the reductions themselves are machine-checked in Lean 4 on top of those certified inputs; the separation between analytic argument and certified input is stated precisely, and tabulated.

1. INTRODUCTION

For $k \geq 1$ put

$$f_k = \sum_{\Omega(n)=k} \frac{1}{n \log n}.$$

Erdős [4] proved that the corresponding sum is bounded uniformly over primitive sets. In particular each f_k converges. Zhang [7] proved $f_k < f_1$ for $k > 1$, and Bayless, Kinlaw and Klyve [8] proved $f_2 > f_3$. Banks and Martin [1] conjectured that

$$f_1 > f_2 > f_3 > \cdots. \tag{1}$$

Lichtman [2] disproved (1), showing that $f_k \to 1$ and that the sequence attains a global minimum at $k = 6$. Gorodetsky, Lichtman and Wong [3] then established

$$f_k = 1 - 2^{-k}\big(a_1 k^2 + O(k \log(k+1))\big), \qquad a_1 = \frac{d \log 2}{4} > 0, \tag{2}$$

where

$$d = \frac{1}{4} \prod_{p \geq 3} \left(1 - \frac{2}{p}\right)^{-1} \left(1 - \frac{1}{p}\right)^{2}. \tag{3}$$

Neither of these yields monotonicity beyond the minimum. A global minimum at $k = 6$ leaves open whether $f_k < f_{k+1}$ holds at every index above it, and the implicit constant in (2) does not make explicit a range on which the increase can be read off. Our first theorem gives that monotonicity in full.

Theorem 1.1. *For every $k \geq 6$ one has $f_k < f_{k+1}$.*

Let

$$g_k = \sum_{\substack{n \text{ odd} \\ \Omega(n)=k}} \frac{1}{n \log n}.$$

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The second theorem shows that deleting the prime 2 reverses the direction of monotonicity at every index.

Theorem 1.2. *For every $k \geq 1$ one has $g_k > g_{k+1}$, and $g_k \to \frac{1}{2}$.*

The proof of Theorem 1.2 gives the more precise expansion

$$g_k = \frac{1}{2} + \beta 2^{-k} + R_k, \qquad |R_k| \leq 4900(k+3)^3 3^{-k} \quad (k \geq 59), \tag{4}$$

where $\gamma = 0.5772156649\ldots$ is Euler's constant and

$$\beta = d\left(\gamma + \log 2 - \sum_{p \geq 3} \frac{\log p}{(p-1)(p-2)}\right) > 0. \tag{5}$$

The 2^{-k} term comes from evaluating a regularized Euler product at $z = 2$. For odd integers the first pole is at $z = 3$, so this evaluation takes place strictly inside the disc of analyticity. For the unrestricted sums the first pole is at $z = 2$ itself. The boundary contribution then acquires the factor k^2 in (2).

The proof has two kinds of input. Identities, coefficient reductions and inequalities are proved in the text. Bounds produced by interval arithmetic are isolated as certified propositions. Thus a sentence of the form "the computation gives" never substitutes for an analytic step: the exact inequality supplied by the computation is stated before it is used.

2. Generating functions and the analytic boundary

For $s > 1$ define

$$P_k(s) = \sum_{\Omega(n)=k} n^{-s}, \qquad P_k^{\mathrm{odd}}(s) = \sum_{\substack{n \text{ odd} \\ \Omega(n)=k}} n^{-s}.$$

Lemma 2.1. *For every $k \geq 1$,*

$$f_k = \int_1^\infty P_k(s)\,\mathrm{d}s, \qquad g_k = \int_1^\infty P_k^{\mathrm{odd}}(s)\,\mathrm{d}s. \tag{6}$$

Proof. For $n \geq 2$,

$$\int_1^\infty n^{-s}\,\mathrm{d}s = \int_1^\infty \mathrm{e}^{-s\log n}\,\mathrm{d}s = \left[-\frac{\mathrm{e}^{-s\log n}}{\log n}\right]_{s=1}^{\infty} = \frac{1}{n\log n}.$$

All summands are non-negative. Tonelli's theorem therefore permits the sum and integral to be interchanged in both cases, giving

$$\int_1^\infty \sum_{\Omega(n)=k} n^{-s}\,\mathrm{d}s = \sum_{\Omega(n)=k} \int_1^\infty n^{-s}\,\mathrm{d}s = f_k,$$

and the same calculation with the sum restricted to odd n proves the second identity. □

The generating functions are

$$\sum_{k\geq 0} P_k(s) z^k = \prod_p (1 - zp^{-s})^{-1}, \qquad \sum_{k\geq 0} P_k^{\mathrm{odd}}(s) z^k = \prod_{p\geq 3} (1 - zp^{-s})^{-1}. \tag{7}$$

Write these products as $\zeta(s)^z H(s,z)$ and $\zeta(s)^z H_{\mathrm{o}}(s,z)$, where

$$H(s,z) = \prod_p (1 - zp^{-s})^{-1}(1-p^{-s})^z, \tag{8}$$

$$H_{\mathrm{o}}(s,z) = (1-2^{-s})^z \prod_{p\geq 3} (1 - zp^{-s})^{-1}(1-p^{-s})^z. \tag{9}$$

Proposition 2.2. *Let $\sigma = \operatorname{Re} s$. Then H is analytic in the two complex variables (s,z) on $\{\sigma > 1/2,\ |z| < 2^\sigma\}$, and H_{o} on $\{\sigma > 1/2,\ |z| < 3^\sigma\}$. In particular, for fixed real $s > 1/2$ the function $H(s,\cdot)$ is analytic in $|z| < 2^s$ and $H_{\mathrm{o}}(s,\cdot)$ in $|z| < 3^s$; each $\partial_s^r H_{\mathrm{o}}(s,\cdot)$ is again analytic in $|z| < 3^s$, and*

$$\frac{\mathrm{d}^r}{\mathrm{d}s^r}[z^j]H_{\mathrm{o}}(s,z) = [z^j]\partial_s^r H_{\mathrm{o}}(s,z); \tag{10}$$

the same holds for H in $|z| < 2^s$.

Proof. Let $x = p^{-s}$, so that $|x| = p^{-\sigma} \leq 2^{-\sigma} < 2^{-1/2}$ on the domain. The factor $(1-x)^z$ is a non-integer power of a complex number, so a branch must be fixed; the principal one will do throughout, since $1-x$ then lies in the disc of radius $2^{-1/2} < 1$ about 1, on which the principal logarithm is analytic and where no branch cut is met. With that convention, for z in a fixed compact set K the logarithm of the pth regularized factor is

$$\begin{aligned} -\log(1-zx) + z\log(1-x) &= \sum_{m\geq 1} \frac{z^m - z}{m} x^m \\ &= \sum_{m\geq 2} \frac{z^m - z}{m} x^m = O_K(p^{-2\sigma}), \end{aligned}$$

the $m = 1$ terms cancelling. The estimate is uniform for $z \in K$ and depends on s only through σ. Since $\sum_p p^{-2\sigma} < \infty$ for $\sigma > 1/2$, the product of the regularized factors converges locally uniformly wherever none of the factors has a pole. The poles occur at $z = p^s$, of modulus p^σ; the nearest has modulus 2^σ for (8) and 3^σ for (9). A locally uniform limit of analytic functions is analytic, which proves the first assertion.

Differentiability in s does not follow from this if s is held real: uniform convergence of real-variable functions says nothing about the derivative of the limit. What is used instead is that the variable s may be taken complex. Each $p^{-s} = \mathrm{e}^{-s\log p}$ is entire in s, so every regularized factor is analytic in (s,z) jointly wherever its denominator does not vanish, which on $\{|z| < 3^\sigma\}$ it does not, since $|z| < 3^\sigma \leq p^\sigma = |p^s|$ for every $p \geq 3$. The estimate above depends on s only through $|p^{-s}| = p^{-\sigma}$, so it holds uniformly on any set with $\sigma \geq s_0 > 1/2$, and $\sum_p p^{-2s_0} < \infty$. The product therefore converges locally uniformly on the stated domain in $\mathbb{C}^2$, and a locally uniform limit of analytic functions of several complex variables is analytic; this is the first assertion. Analyticity in s gives derivatives of every order at real s, and since $[z^j]$ is a Cauchy integral over a circle that may be held fixed on a compact subset, the two operations commute, which is (10). □

At $s = 1$ the point $z = 2$ lies on the first singular circle for H and lies strictly inside the first singular circle for H_{o}. This is the analytic distinction used below.

We shall also need the local change of variables used in the odd case.

Lemma 2.3. *Let $u = \log\zeta(s)$ for $s > 1$, and let $s = s(u)$ be the inverse function. Put $\varepsilon(u) = s(u) - 1$ and $J(u) = -s'(u)$. As $u \to \infty$,*

$$\varepsilon(u) = \mathrm{e}^{-u} + \gamma\mathrm{e}^{-2u} + O(\mathrm{e}^{-3u}), \qquad J(u) = \mathrm{e}^{-u} + 2\gamma\mathrm{e}^{-2u} + O(\mathrm{e}^{-3u}). \tag{11}$$

Proof. The point of the argument is that the expansions below are convergent power series and not merely asymptotic, which is what allows the second of them to be differentiated term by term.

Since ζ is meromorphic on $\mathbb{C}$ with a single simple pole, of residue 1, at $s = 1$, the function $\varepsilon \mapsto \varepsilon\,\zeta(1+\varepsilon)$ is analytic near $\varepsilon = 0$ with value 1 there; its expansion is $1 + \gamma\varepsilon + O(\varepsilon^2)$. Hence

$$x = \frac{1}{\zeta(1+\varepsilon)} = \frac{\varepsilon}{\varepsilon\,\zeta(1+\varepsilon)}$$

is analytic near $\varepsilon = 0$, with $x(0) = 0$ and $x = \varepsilon - \gamma\varepsilon^2 + O(\varepsilon^3)$, so in particular $\mathrm{d}x/\,\mathrm{d}\varepsilon = 1 \neq 0$ at the origin. By the inverse function theorem for analytic functions, ε is therefore an analytic function of x near $x = 0$, given by a convergent power series

$$\varepsilon = \psi(x) = x + \gamma x^2 + O(x^3),$$

which is the first formula in (11) once $x = \mathrm{e}^{-u}$. Because ψ is analytic rather than merely asymptotic, it may be differentiated termwise: $\psi'(x) = 1 + 2\gamma x + O(x^2)$. With $x = \mathrm{e}^{-u}$ one has $\mathrm{d}x/\,\mathrm{d}u = -x$, so the chain rule gives

$$\varepsilon'(u) = \psi'(x)\frac{\mathrm{d}x}{\mathrm{d}u} = -x\big(1 + 2\gamma x + O(x^2)\big) = -x - 2\gamma x^2 + O(x^3),$$

and $J(u) = -\varepsilon'(u) = x + 2\gamma x^2 + O(x^3)$. □

3. The odd sums

Throughout this section write

$$h_j(s) = [z^j]H_{\mathrm{o}}(s,z), \qquad b_j = h_j'(1) + 2\gamma h_j(1).$$

Since $\zeta(s)^z = \sum_{m\geq 0}(\log\zeta(s))^m z^m/m!$, coefficient extraction in (7) gives the exact identity

$$P_k^{\mathrm{odd}}(s) = \sum_{j=0}^{k} h_j(s)\frac{(\log\zeta(s))^{k-j}}{(k-j)!}. \tag{12}$$

Using $u = \log\zeta(s)$ in Lemma 2.1, and noting that s decreases from ∞ to 1 as u increases from 0 to ∞, we obtain

$$g_k = \sum_{j=0}^{k}\int_0^{\infty} \frac{u^{k-j}}{(k-j)!}\,h_j(s(u))J(u)\,\mathrm{d}u. \tag{13}$$

3.1. The first two terms.

Proposition 3.1. *With E_j, h_j and b_j as in the proof below, put*

$$R_k = \sum_{j=0}^{k}\int_0^{\infty}\frac{u^{k-j}}{(k-j)!}E_j(u)\,\mathrm{d}u - \sum_{j>k} h_j(1) - 2^{-k-1}\sum_{j>k} b_j 2^j. \tag{14}$$

Then

$$g_k = H_{\mathrm{o}}(1,1) + 2^{-k-1}\big(\partial_s H_{\mathrm{o}}(1,2) + 2\gamma H_{\mathrm{o}}(1,2)\big) + R_k. \tag{15}$$

Proof. Let $x = \mathrm{e}^{-u}$. By Lemma 2.3, $J(u) = x + 2\gamma x^2 + O(x^3)$ and $\varepsilon(u)J(u) = x^2 + O(x^3)$. Taylor's theorem at $s = 1$ gives

$$h_j(1+\varepsilon) = h_j(1) + \varepsilon h_j'(1) + \frac{\varepsilon^2}{2}h_j''(1+\theta_{j,u}\varepsilon)$$

for some $0 < \theta_{j,u} < 1$. Put

$$r_0(u) = J(u) - \mathrm{e}^{-u} - 2\gamma\mathrm{e}^{-2u}, \qquad r_1(u) = \varepsilon(u)J(u) - \mathrm{e}^{-2u}.$$

Multiplication by $J(u)$ now gives the exact identity

$$\begin{aligned} h_j(s(u))J(u) &= h_j(1)\mathrm{e}^{-u} + b_j\mathrm{e}^{-2u} + E_j(u), \\ E_j(u) &= h_j(1)r_0(u) + h_j'(1)r_1(u) + \frac{\varepsilon(u)^2 J(u)}{2} h_j''(1 + \theta_{j,u}\varepsilon(u)). \end{aligned} \tag{16}$$

Thus no error is suppressed in the notation E_j: its three summands are the Jacobian error, the first-order coordinate error and the Taylor remainder.

Insert (16) into (13). Since

$$\int_0^\infty \frac{u^m}{m!}\mathrm{e}^{-ru}\,\mathrm{d}u = r^{-m-1} \qquad (r > 0),$$

the first two terms become

$$\sum_{j=0}^{k} h_j(1) = H_{\mathrm{o}}(1,1) - \sum_{j>k} h_j(1), \tag{17}$$

$$\sum_{j=0}^{k} b_j 2^{-(k-j+1)} = 2^{-k-1}\big(\partial_s H_{\mathrm{o}}(1,2) + 2\gamma H_{\mathrm{o}}(1,2)\big) - 2^{-k-1}\sum_{j>k} b_j 2^j. \tag{18}$$

Substituting (17) and (18), and collecting the E_j integrals with the two tails, is exactly (14); this gives (15). Note that (14) is an identity, not an error term absorbed into an O: every later estimate is an estimate of those three explicit quantities. □

Lemma 3.2. *One has*

$$H_{\mathrm{o}}(1,1) = \frac{1}{2}, \qquad H_{\mathrm{o}}(1,2) = d, \tag{19}$$

and

$$\partial_s \log H_{\mathrm{o}}(1,2) = 2\log 2 - 2\sum_{p\geq 3} \frac{\log p}{(p-1)(p-2)}. \tag{20}$$

Proof. At $s = z = 1$, every factor in the product over $p \geq 3$ cancels, and hence

$$H_{\mathrm{o}}(1,1) = (1 - 2^{-1}) = \frac{1}{2}.$$

At $(s,z) = (1,2)$,

$$H_{\mathrm{o}}(1,2) = \frac{1}{4}\prod_{p\geq 3}\left(1 - \frac{2}{p}\right)^{-1}\left(1 - \frac{1}{p}\right)^2 = d.$$

For the derivative, take logarithms:

$$\log H_{\mathrm{o}}(s,2) = 2\log(1 - 2^{-s}) + \sum_{p\geq 3}\left[-\log(1 - 2p^{-s}) + 2\log(1 - p^{-s})\right].$$

The differentiated pth summand is $O((\log p)p^{-2s})$ locally uniformly near $s = 1$, so termwise differentiation is permitted. At $s = 1$ the first term contributes $2\log 2$, while the pth summand contributes

$$-\frac{2\log p}{p-2} + \frac{2\log p}{p-1} = -\frac{2\log p}{(p-1)(p-2)}.$$

Summing over $p \geq 3$ proves (20). □

Combining Proposition 3.1 and Lemma 3.2 gives

$$g_k = \frac{1}{2} + \beta 2^{-k} + R_k, \qquad \beta = d\left(\gamma + \log 2 - \sum_{p\geq 3} \frac{\log p}{(p-1)(p-2)}\right). \tag{21}$$

3.2. **The remainder.** Three weighted ℓ^1 norms of coefficients carry all of the interval input for the odd sums. Put

$$\begin{aligned} \Sigma_0 &= \sum_{m\geq 0} 3^m |[z^m](1-z/3)H_{\mathrm{o}}(1,z)|, \\ \Sigma_1 &= \sum_{m\geq 0} 3^m |[z^m](1-z/3)^2 \partial_s H_{\mathrm{o}}(1,z)|, \\ \Sigma_2 &= \sup_{1\leq s\leq s(3)} \sum_{m\geq 0} 3^m |[z^m](1-z/3)^3 \partial_s^2 H_{\mathrm{o}}(s,z)|. \end{aligned} \tag{22}$$

Finiteness of Σ_0 and Σ_1 is analytic, and the weights $r = 1, 2$ on the factors $(1-z/3)^r$ are what makes it so. Proposition 2.2 gives $H_{\mathrm{o}}(1,\cdot)$ only on $|z| < 3$, which by itself would not make a sum weighted by 3^m converge. But at $s = 1$ the nearest pole in z is at $z = 3$, contributed by the single Euler factor $(1-z/3)^{-1}$, and after $r-1$ derivatives in s that pole has order r; multiplying by $(1-z/3)^r$ cancels it exactly, so both functions continue analytically to $|z| < 5$, the next pole being at $z = 5$. Their coefficients decay like 5^{-m} and the weighted sums converge geometrically.

For Σ_2 the same device applies only at the left endpoint. At $s > 1$ the pole sits at $z = 3^s$, which $(1-z/3)^3$ does not cancel, and the inner sum is finite because the radius of convergence is then $3^s > 3$ — a margin that shrinks to nothing as $s \to 1^+$, so this does not by itself bound the supremum. Each inner sum is finite for every s in the range, by cancellation at $s = 1$ and by $3/3^s < 1$ for $s > 1$; the finiteness of the supremum over $[1, s(3)]$ is part of what the interval computation below establishes, not a separate analytic claim. For Σ_0 and Σ_1 only the numerical values are certified.

Next, split the remainder of Proposition 3.1 at $u = 3$: with E_j the explicit expression in (16), put

$$R_{k,\mathrm{far}} = \sum_{j=0}^{k} \int_3^\infty \frac{u^{k-j}}{(k-j)!} E_j(u)\,\mathrm{d}u, \qquad R_{k,\mathrm{near}} = \sum_{j=0}^{k} \int_0^3 \frac{u^{k-j}}{(k-j)!} E_j(u)\,\mathrm{d}u, \tag{23}$$

so that R_k is $R_{k,\mathrm{far}} + R_{k,\mathrm{near}}$ together with the two coefficient tails of (17)–(18). Here $s(3)$ denotes the value of s at $u = 3$.

The far range needs one further constant. Write $\varepsilon = s - 1$ and $J = \zeta(s)/(-\zeta'(s))$; this is the J of Lemma 2.3 expressed in s rather than u, since $\mathrm{d}u/\,\mathrm{d}s = \zeta'(s)/\zeta(s)$ makes $-s'(u)$ equal to $\zeta(s)/(-\zeta'(s))$, and the four quantities below compare it with powers of $\zeta^{-1} = \mathrm{e}^{-u}$. Let

$$\begin{gathered} C_b = \sup|\varepsilon\zeta|, \qquad C_f = \sup|\zeta^2\big(J - \zeta^{-1}\big)|, \\ C_J = \sup\big|\zeta^3\big(J - \zeta^{-1} - 2\gamma\zeta^{-2}\big)\big|, \qquad C_P = \sup\big|\zeta^3\big(\varepsilon J - \zeta^{-2}\big)\big|, \\ C_E = C_J + C_P + C_b^2\,(C_f + 1), \end{gathered} \tag{24}$$

every supremum being over $1 < s \leq s(3)$. Each of the four is a ratio in which the pole of ζ at $s = 1$ cancels. How they are computed matters, and is described in Section 8: evaluating them on boxes of s loses that cancellation and the enclosures diverge however finely the range is partitioned, so a Taylor model is used instead.

Proposition 3.3 (Certified odd input)**.** *With Σ_i as in* (22) *and C_E as in* (24)*, the following are certified by interval arithmetic.*
(i) The coefficient norms satisfy

$$\Sigma_0 \le 44.5987, \qquad \Sigma_1 \le 486.1945, \qquad \Sigma_2 \le 14414, \qquad C_E \le 5.48910. \tag{25}$$

(ii) On the far range,

$$\sum_{j=0}^{k} \int_3^\infty \frac{u^{k-j}}{(k-j)!} |E_j(u)| \, \mathrm{d}u \le \frac{C_E \Sigma_2}{18} (k+3)^3 3^{-k}. \tag{26}$$

(iii) On the near range the two groups of terms in (16) *are bounded separately:*

$$\sum_{j=0}^{k} \int_0^3 \frac{u^{k-j}}{(k-j)!} |h_j(s(u))| J(u) \, \mathrm{d}u \le C_N 3^{-k} \quad (59 \le k \le 200), \quad C_N \le 367.8435, \tag{27}$$

$$\sum_{j=0}^{k} \int_0^3 \frac{u^{k-j}}{(k-j)!} \big(|h_j(1)| \mathrm{e}^{-u} + |b_j| \mathrm{e}^{-2u} \big) \, \mathrm{d}u \le N_U (k+3)^3 3^{-k}, \qquad N_U \le 26.0962. \tag{28}$$

(iv) $\beta > 0$, and

$$4900 (62)^3 3^{-59} < \frac{\beta}{3} 2^{-59}. \tag{29}$$

(v) Direct interval evaluation gives $g_k > g_{k+1}$ for $1 \le k \le 75$.

Part (v) is not a coefficient bound but a sweep, and it is what makes the range restriction in (27) harmless.

That restriction is a feature of the method rather than of the inequality. The h_j alternate in sign, so the left-hand side has to be majorized rather than evaluated in closed form, and the majorant used in the proof of Proposition 3.5 is at its loosest where $a(s) = 2|h_1(s)|$ is largest, which is at small k. The bound it yields exceeds $C_N 3^{-k}$ for $8 \le k \le 25$, peaking at $724.55 \cdot 3^{-k}$ at $k = 14$, and settles to $34.14 \cdot 3^{-k}$ by $k = 59$. We make no claim about the left-hand side itself outside $k \ge 59$: it is the certified bound, not the quantity, that needs the restriction, and $k \ge 26$ would in fact serve. We state $k \ge 59$ because that is the only range in which (31) is ever applied, and monotonicity below it comes from (v) without appeal to (31).

The numeral $C_N \le 367.8435$ is carried, rather than the 34.14 actually computed on $k \ge 59$, so that the ledger below is unchanged. It is not idle either: 367.8435 is the supremum over all $k \ge 1$ of $3^k P_k^{\mathrm{odd}}(s(3))$, the signed quantity whose coefficient identity earlier drafts of this proof used, attained at $k = 11$.

Lemma 3.4. *Let $A(z) = \sum_{j \ge 0} a_j z^j$.*
(i) If $\sum_m 3^m |[z^m](1 - z/3)A(z)| \le S$, then $|a_j| \le S 3^{-j}$.
(ii) If $\sum_m 3^m |[z^m](1 - z/3)^2 A(z)| \le S$, then $|a_j| \le S(j+1)3^{-j}$.
(iii) If $\sum_m 3^m |[z^m](1 - z/3)^3 A(z)| \le S$, then

$$\sum_{j=0}^{k} 3^j |a_j| \le \binom{k+3}{3} S. \tag{30}$$

Proof. Write $(1 - z/3)^r A(z) = \sum_m w_m z^m$. Then

$$A(z) = \left(\sum_{q \ge 0} \binom{q+r-1}{r-1} 3^{-q} z^q \right) \left(\sum_{m \ge 0} w_m z^m \right).$$

For $r=1$ this gives $|a_j| \le 3^{-j}\sum_{m\le j} 3^m|w_m| \le S3^{-j}$. For $r=2$ it gives $|a_j| \le (j+1)3^{-j}S$. For $r=3$,

$$\begin{aligned}\sum_{j=0}^{k} 3^j|a_j| &\le \sum_{m=0}^{k} 3^m|w_m| \sum_{q=0}^{k-m} \binom{q+2}{2} \\ &= \sum_{m=0}^{k} 3^m|w_m| \binom{k-m+3}{3} \le \binom{k+3}{3} S.\end{aligned}$$

□

Proposition 3.5. *For every $k \ge 59$,*

$$|R_k| \le 4900(k+3)^3 3^{-k}. \tag{31}$$

The threshold is not an artefact of the method but of the near range: the coefficients h_j alternate in sign, so the near-range term has to be estimated through the majorant (34), and the resulting bound exceeds (31) for small k. Nothing is lost, since monotonicity for $k \le 75$ is certified directly and does not use (31); see the proof of Theorem 1.2.

Proof. Write R_k as the sum of the far contribution, the near contribution, and the two coefficient tails of (17)–(18). The far term and the near-range main term are cubic in k; the other two are bounded by constant multiples of 3^{-k}, and since $(k+3)^3 \ge 1$ the cubic factor may be absorbed once, at the end.

Far range $u \ge 3$. The exact error is the sum of the three terms of (16)—no asymptotic series is summed—so the certified Taylor model of (26) applies to the integrand. Its k-dependence is supplied not by that computation, which certifies finitely many numbers, but by Lemma 3.4(iii): applied to the second s-derivative of H_{o} with $S = \Sigma_2$, (30) gives $\sum_{j\le k} 3^j|\partial_s^2 h_j| \le \binom{k+3}{3}\Sigma_2 \le (k+3)^3\Sigma_2/6$, and the remaining factor 3 comes from the width of the u-range in (26). Hence $|R_{k,\mathrm{far}}| \le \frac{C_E\Sigma_2}{18}(k+3)^3 3^{-k}$, the $18 = 6\cdot 3$ recording those two steps.

First tail. By Lemma 3.4(i), $\left|\sum_{j>k} h_j(1)\right| \le \Sigma_0 \sum_{j>k} 3^{-j} = \frac{\Sigma_0}{2} 3^{-k}$.

Second tail. Put $B(z) = \partial_s H_{\mathrm{o}}(1,z) + 2\gamma H_{\mathrm{o}}(1,z)$. Multiply by $(1-z/3)^2$ and treat the two summands separately. The first contributes Σ_1 by its definition in Proposition 3.3. For the second, write $(1-z/3)^2 H_{\mathrm{o}} = (1-z/3)\cdot\big[(1-z/3)H_{\mathrm{o}}\big]$; the weighted ℓ^1 norm $\sum_m 3^m|[z^m]\cdot|$ is submultiplicative under this extra factor $(1-z/3)$, whose own norm is 2, so the second summand contributes at most $2\gamma\cdot 2\Sigma_0 = 4\gamma\Sigma_0$ — this is where the 4 comes from, rather than the 2 of the prefactor. Since $\gamma < 0.578$,

$$\sum_{m\ge 0} 3^m|[z^m](1-z/3)^2 B(z)| \le \Sigma_1 + 4\gamma\Sigma_0 < 590,$$

so $|b_j| \le 590(j+1)3^{-j}$ by Lemma 3.4(ii). Summing the resulting geometric series,

$$2^{-k-1}\sum_{j>k}|b_j|2^j \le 590\cdot 2^{-k-1}\sum_{j>k}(j+1)(2/3)^j = 590(k+4)3^{-k},$$

and $(k+4) \le \frac{5}{64}(k+3)^3$ for $k \ge 1$, with equality at $k=1$.

Near range $0 < u \le 3$. Here $|E_j|$ is at most the true integrand plus the two subtracted main terms, and the latter contribute at most $N_U(k+3)^3 3^{-k}$ by (28). For the former, (12) and the substitution $s = s(u)$, under which $u = \log\zeta(s)$ and $J(u)\,\mathrm{d}u = \mathrm{d}s$, give

$$\sum_{j=0}^{k}\int_0^3 \frac{u^{k-j}}{(k-j)!}|h_j(s(u))|J(u)\,\mathrm{d}u = \int_{s(3)}^{\infty} A_k(s)\,\mathrm{d}s, \qquad A_k(s) = \sum_{j=0}^{k}\frac{(\log\zeta(s))^{k-j}}{(k-j)!}|h_j(s)|. \tag{32}$$

It is A_k, and not the signed sum, that has to be estimated: writing $\Pi(t) = \sum_{p\geq 3} p^{-t}$, the z-linear coefficient of $\log H_{\mathrm{o}}$ in (9) is

$$h_1(s) = \Pi(s) - \log\zeta(s) = \log(1-2^{-s}) - \sum_{m\geq 2} \frac{\Pi(ms)}{m} < 0, \tag{33}$$

both terms being negative, so the h_j are not of one sign and the identity $\sum_j u^{k-j} h_j(s)/(k-j)! = P_k^{\mathrm{odd}}(s)$ bounds A_k from below rather than above.

We majorize instead. By (33) every coefficient of $\log H_{\mathrm{o}}(s,z) = h_1(s)z + \sum_{m\geq 2}\Pi(ms)z^m/m$ beyond the first is positive, so replacing h_1 by $|h_1|$ produces

$$H_{\mathrm{o}}^+(s,z) = H_{\mathrm{o}}(s,z)\mathrm{e}^{-2h_1(s)z},$$

a power series with non-negative coefficients dominating $|h_j|$ termwise. Hence, with $a(s) = 2|h_1(s)|$,

$$A_k(s) \leq [z^k]\big(\zeta(s)^z H_{\mathrm{o}}^+(s,z)\big) = \sum_{i=0}^{k} P_{k-i}^{\mathrm{odd}}(s)\frac{a(s)^i}{i!} =: P_k^+(s). \tag{34}$$

To estimate $\int_{s(3)}^{\infty} P_k^+$, fix a cut point $s_0 > s(3)$ and treat the two pieces differently. Each is effective for its own reason.

On the compact piece $[s(3), s_0]$, every P_j^{odd} is a Dirichlet series with positive terms and, by (33), a is a positive combination of such series; a sum of products of non-negative decreasing functions is decreasing, so P_k^+ decreases in s. A finite left-endpoint Riemann sum over $[s(3), s_0]$ is therefore an upper bound for $\int_{s(3)}^{s_0} P_k^+$, with no quadrature error to estimate — which is what makes this step certifiable without any analysis of the quadrature rule itself.

On the tail $[s_0, \infty)$ the integral converges at a certified rate. Every odd n with $\Omega(n) = k-i$ satisfies $n \geq 3^{k-i}$, so for $s \geq s_0$, writing $t = s - s_0$, we get $P_{k-i}^{\mathrm{odd}}(s) \leq 3^{-(k-i)t} P_{k-i}^{\mathrm{odd}}(s_0)$, while every Dirichlet term of a has $n \geq 2$ and hence $a(s) \leq 2^{-t} a(s_0)$ by (33). Since $3^{-x} \leq 2^{-x}$ for $x \geq 0$, the ith term of P_k^+ is multiplied by at most $3^{-(k-i)t}2^{-it} \leq 2^{-kt}$; all terms being non-negative, Tonelli permits termwise integration, and $\int_0^{\infty} 2^{-kt}\,\mathrm{d}t = 1/(k\log 2)$ gives

$$\int_{s_0}^{\infty} P_k^+(s)\,\mathrm{d}s \leq \frac{P_k^+(s_0)}{k\log 2}. \tag{35}$$

Adding the Riemann sum over $[s(3), s_0]$ to (35) bounds $\int_{s(3)}^{\infty} P_k^+$, hence $\int_{s(3)}^{\infty} A_k$ by (34). For one k this is a finite computation, and (27) is its outcome for each of the 142 indices $59 \leq k \leq 200$.

That leaves the dependence on k, which no finite computation can settle, and which has to be reduced exactly as the far range's was. Here the reduction is Cauchy's estimate. Because $H_{\mathrm{o}}^+(s,\cdot)$ has non-negative coefficients, its maximum on a circle $|z| = R$ is attained at $z = R$, so for any $R < 3^{s(3)}$ and any $s \geq s(3)$,

$$A_k(s) \leq R^{-k}\zeta(s)^R H_{\mathrm{o}}(s,R)\mathrm{e}^{a(s)R} = R^{-k}\prod_{p\geq 3}\big(1-Rp^{-s}\big)^{-1}\mathrm{e}^{a(s)R}, \tag{36}$$

a bound whose only k-dependence is the factor R^{-k}. It cannot be integrated over the whole of $[s(3), \infty)$, since its right-hand side tends to 1 rather than to 0 as $s \to \infty$; so we use it on $[s(3), s_1]$ for a cut point s_1 and fall back on the decay already established beyond it. On $[s(3), s_1]$, (36) gives $\int_{s(3)}^{s_1} A_k \leq R^{-k}\Theta_1$ with $\Theta_1 = \int_{s(3)}^{s_1}\prod_{p\geq 3}(1-Rp^{-s})^{-1}\mathrm{e}^{a(s)R}\,\mathrm{d}s$, finite and independent of k. Beyond s_1, (35) applies with $s_0 = s_1$, and bounding $P_k^+(s_1)$ by (36) in turn

gives $\int_{s_1}^{\infty} A_k \le R^{-k}\Theta_2/(k\log 2)$ with $\Theta_2 = \prod_{p\ge 3}(1-Rp^{-s_1})^{-1}\mathrm{e}^{a(s_1)R}$, again independent of k. Hence

$$3^k \int_{s(3)}^{\infty} A_k \le \Big(\frac{3}{R}\Big)^k \Big(\Theta_1 + \frac{\Theta_2}{k\log 2}\Big),$$

and with $R = 3.15$, which is below $3^{s(3)} = 3.1727\ldots$, the factor $(3/3.15)^{201} < 5.6\cdot 10^{-5}$ puts the right-hand side below 10 for every $k > 200$. Therefore

$$\int_{s(3)}^{\infty} A_k(s)\,\mathrm{d}s \le C_N 3^{-k} \qquad \text{for all } k \ge 59, \tag{37}$$

which is the range in which (31) is used. The two branches meet at $k = 200$ with room to spare: the largest value over the finite range is $34.14\cdot 3^{-k}$ and the Cauchy branch gives at most $9.66\cdot 3^{-k}$, against the $C_N = 367.8435$ carried in the ledger.

The companion bound (28) reduces the same way. The first-order pole bounds $|h_j(1)| \le \Sigma_0 3^{-j}$ and $|b_j| \le (j+1)\Sigma_1 3^{-j}$ of Lemma 3.4(i)–(ii) turn each integral into a regularized incomplete gamma value, and $P(m,x) \le x^m/m!$ makes the resulting sums telescope: the first is at most $\Sigma_0 \mathrm{e}^9 3^{-k}$ and the second at most $\frac{1}{2}\big((k+1)\Sigma_1 + 2\gamma\Sigma_0\big)\mathrm{e}^9 3^{-k}$. Both are of lower order than $(k+3)^3 3^{-k}$, so the ratio to the budget decreases in k and the maximum over a finite head governs; it is taken over $1 \le k \le 400$.

Substituting the certified constants, the four contributions are

$$|R_k| \le \Big(\underbrace{\frac{5.48910\cdot 14414}{18}}_{4395.55} + \underbrace{\frac{44.5987}{2} + \frac{5\cdot 590}{64} + C_N + N_U}_{462.34}\Big)(k+3)^3 3^{-k} < 4900(k+3)^3 3^{-k},$$

with $C_N = 367.8435$ and $N_U = 26.0962$. □

Proof of Theorem 1.2. The limit follows at once from (21) and Proposition 3.5. For monotonicity,

$$g_k - g_{k+1} = \beta 2^{-k-1} + R_k - R_{k+1}. \tag{38}$$

It is enough to prove $|R_m| < \beta 2^{-m}/3$ for $m = k, k+1$, because then

$$|R_k| + |R_{k+1}| < \frac{2\beta}{3}2^{-k-1} + \frac{\beta}{3}2^{-k-1} = \beta 2^{-k-1}.$$

Define

$$q(k) = \frac{3\cdot 4900}{\beta}(k+3)^3(2/3)^k.$$

The certified inequality (29) says $q(59) < 1$. Moreover

$$\frac{q(k+1)}{q(k)} = \frac{2}{3}\left(\frac{k+4}{k+3}\right)^3 < 1 \qquad (k \ge 4).$$

Thus $q(k) < 1$ for every $k \ge 59$, and Proposition 3.5 gives $|R_k| < \beta 2^{-k}/3$ throughout that range. Equation (38) proves monotonicity for $k \ge 59$. The certified direct computation covers $1 \le k \le 75$, so the two ranges cover every $k \ge 1$. They overlap on the seventeen indices $59 \le k \le 75$. □

4. The unrestricted sums: an exact criterion

Write

$$f_k = 1 - 2^{-k}\big(a_1 k^2 + E(k)\big), \qquad a_1 = \frac{d\log 2}{4}. \tag{39}$$

Then

$$f_{k+1} - f_k = 2^{-k-1}\left(a_1(k^2 - 2k - 1) + 2E(k) - E(k+1)\right). \tag{40}$$

Thus $f_k < f_{k+1}$ if and only if

$$a_1(k^2 - 2k - 1) + 2E(k) - E(k+1) > 0. \tag{41}$$

We now derive the decomposition of $E(k)$ used to bound the second term.

Set

$$F(s,z) = \prod_p (1 - zp^{-s})^{-1}, \qquad G_2(s,z) = (s-1)^z(1 - z2^{-s})F(s,z),$$

and write $c_i = [z^i]G_2(1,z)$. Since $G_2(1,z) = H_{\mathrm{o}}(1,z)$, the power series has radius 3 and

$$\sum_{i\ge 0} c_i = \frac{1}{2}, \qquad \sum_{i\ge 0} 2^i c_i = d. \tag{42}$$

The coefficients are those of Section 3:

$$c_i = h_i(1). \tag{43}$$

Define the absolutely convergent signed moments

$$U = \sum_{i\ge 0} i 2^i c_i, \qquad V = \sum_{i\ge 0} i^2 2^i c_i. \tag{44}$$

Following [3, §3], put

$$I_k = \int_1^2 [z^k]F(s,z)\,\mathrm{d}s, \tag{45}$$

$$I'_k = \sum_{i+j+\ell=k} c_i \int_1^2 \frac{(-\log(s-1))^\ell}{\ell!} 2^{-js}\,\mathrm{d}s. \tag{46}$$

For $m \ge 0$ define

$$A_m = 2^m \sum_{j+\ell=m} \int_1^2 \frac{(-\log(s-1))^\ell}{\ell!} 2^{-js}\,\mathrm{d}s \tag{47}$$

and

$$\delta_m = A_m - 2^{m+1} + \frac{\log 2}{4} m^2, \qquad \Delta(k) = \sum_{i=0}^{k} 2^i c_i \delta_{k-i}. \tag{48}$$

Proposition 4.1. *Let*

$$E_{S_1}(k) = -2^k(I_k - I'_k), \tag{49}$$

$$L_k = 2\sum_{i>k} c_i, \qquad Q_k = \sum_{i>k} 2^i c_i (k-i)^2, \tag{50}$$

$$T_k = f_k - I_k, \tag{51}$$

$$E_{\exp}(k) = 2^k L_k - \frac{\log 2}{4} Q_k - 2^k T_k. \tag{52}$$

Then the identity

$$E(k) = E_{S_1}(k) + \frac{\log 2}{4}(V - 2kU) - \Delta(k) + E_{\exp}(k) \tag{53}$$

holds for every $k \geq 1$.

Proof. From (46) and (47),

$$I'_k = 2^{-k} \sum_{i=0}^{k} 2^i c_i A_{k-i}.$$

Insert the definition of δ_{k-i}:

$$I'_k = 2 \sum_{i=0}^{k} c_i - \frac{\log 2}{4} 2^{-k} \sum_{i=0}^{k} 2^i c_i (k-i)^2 + 2^{-k}\Delta(k). \tag{54}$$

By (42) and (44),

$$2 \sum_{i=0}^{k} c_i = 1 - L_k,$$

$$\begin{aligned} \sum_{i=0}^{k} 2^i c_i (k-i)^2 &= \sum_{i \geq 0} 2^i c_i (k-i)^2 - Q_k \\ &= dk^2 - 2kU + V - Q_k. \end{aligned}$$

Since $a_1 = \frac{\log 2}{4} d$, substitution into (54) gives

$$I'_k = 1 - 2^{-k}\left(a_1 k^2 + \frac{\log 2}{4}(V - 2kU) - \Delta(k) + 2^k L_k - \frac{\log 2}{4} Q_k \right).$$

Finally $f_k = I'_k + (I_k - I'_k) + T_k$. Comparing with (39) yields (53). □

5. Bounds for the four source terms

5.1. **The S_1 term.** Set

$$\alpha_i = \sup_{1 \leq t \leq 2} |[z^i] \partial_s G_2(t, z)|, \qquad \alpha_+ = \frac{1}{2} \sum_{i \geq 0} \alpha_i 2^i.$$

The series converges, and a pole location at one value of t is not enough to see it: what is needed is a bound on the coefficients uniform in t.

First, G_2 is analytic in s on a neighbourhood of $[1, 2]$, the apparent branch point of $(s-1)^z$ at $s = 1$ notwithstanding. Indeed $F(s, z) = \zeta(s)^z H(s, z)$ by (8), so

$$G_2(s, z) = \big((s-1)\zeta(s)\big)^z (1 - z2^{-s}) H(s, z),$$

and $(s-1)\zeta(s)$ is analytic near $s = 1$ with value 1 there, the pole of ζ being simple with residue 1; hence $\big((s-1)\zeta(s)\big)^z = \exp\big(z \log((s-1)\zeta(s))\big)$ is analytic there too, and H is analytic by Proposition 2.2. (At $s = 1$ this also gives $G_2(1, z) = H_{\mathrm{o}}(1, z)$, as used above.)

The first z-pole of $\partial_s G_2(t, \cdot)$ is at $z = 3^t$, so for $t \in [1, 2]$ all of these functions are analytic in $|z| < 3$. Fix r with $2 < r < 3$. Then $M = \sup_{1 \leq t \leq 2} \sup_{|z|=r} |\partial_s G_2(t, z)|$ is finite, the set $[1, 2] \times \{|z| = r\}$ being compact and $\partial_s G_2$ continuous on it by the previous paragraph, and Cauchy's estimate on the fixed circle $|z| = r$ gives $\alpha_i \leq M r^{-i}$ uniformly in t. Hence $\sum_{i \geq 0} \alpha_i 2^i \leq M(1 - 2/r)^{-1} < \infty$.

Lemma 5.1. *For every $k \geq 1$,*

$$|E_{S_1}(k)| \leq \alpha_+(k+1), \tag{55}$$

and hence

$$|2E_{S_1}(k) - E_{S_1}(k+1)| \leq \alpha_+(3k+4). \tag{56}$$

Proof. The mean value theorem gives

$$|[z^i]G_2(s,z) - c_i| \leq (s-1)\alpha_i.$$

Subtracting (46) from the coefficient expansion for I_k and then setting $s = 1 + \mathrm{e}^{-t}$ gives

$$\begin{aligned} |I_k - I_k'| &\leq \sum_{i+j+\ell=k} \alpha_i \int_0^\infty \frac{t^\ell}{\ell!} 2^{-j(1+\mathrm{e}^{-t})} \mathrm{e}^{-2t}\,\mathrm{d}t \\ &\leq \sum_{i+j+\ell=k} \alpha_i 2^{-j} \int_0^\infty \frac{t^\ell}{\ell!} \mathrm{e}^{-2t}\,\mathrm{d}t \\ &= \sum_{i+j+\ell=k} \alpha_i 2^{-j-\ell-1}. \end{aligned}$$

Multiplying by 2^k and using $k = i + j + \ell$,

$$|E_{S_1}(k)| \leq \frac{1}{2}\sum_{i=0}^{k} \alpha_i 2^i (k-i+1) \leq \alpha_+(k+1).$$

The triangle inequality now gives

$$|2E_{S_1}(k) - E_{S_1}(k+1)| \leq 2\alpha_+(k+1) + \alpha_+(k+2) = \alpha_+(3k+4).$$

□

5.2. **The signed moment term.** The combination of signed moments that has to be bounded is not chosen; it is what the difference of two consecutive source terms produces:

$$2(V - 2kU) - \big(V - 2(k+1)U\big) = V + (2-2k)U. \tag{57}$$

Since this is an identity, bounding the right-hand side is an exact optimization over the rectangle of admissible (U, V), not a termwise absolute-value estimate. That is the reason for evaluating the moments before absolute values are taken: replacing them by $\sum_i 2^i|c_i|$ would discard the cancellation in (57) and cost a factor in k.

Lemma 5.2. *If $|U| \leq U_{\max}$ and $|V| \leq V_{\max}$, then for $k \geq 2$,*

$$\frac{\log 2}{4}\big(V + (2-2k)U\big) \geq -\frac{\log 2}{4}\big(V_{\max} + (2k-2)U_{\max}\big). \tag{58}$$

Proof. The coefficient of V is 1, while the coefficient $2-2k$ of U is negative. The minimum over the rectangle $[-U_{\max}, U_{\max}] \times [-V_{\max}, V_{\max}]$ is therefore attained at $U = U_{\max}$ and $V = -V_{\max}$, giving (58). □

5.3. **The δ term.** The next lemma supplies the self-contained bound used for $\Delta(k)$.

Lemma 5.3. *For $j \geq 0$ and $u \geq 0$ put*

$$B_j(u) = 2^{-j\mathrm{e}^{-u}} - 1 + j \log 2\, \mathrm{e}^{-u}$$

and

$$\eta_i(j) = \frac{1}{i!} \int_0^\infty u^i \mathrm{e}^{-u} B_j(u)\, \mathrm{d}u.$$

Then $B_j(u) \geq 0$, $\eta_i(j) \geq 0$, and

$$\delta_m = -1 - \frac{\log 2}{4} m + \sum_{i=0}^{m} 2^i \eta_i(m-i). \tag{59}$$

Proof. With $s = 1 + \mathrm{e}^{-u}$ in (47),

$$A_m = \sum_{i=0}^{m} \frac{2^i}{i!} \int_0^\infty u^i \mathrm{e}^{-u} 2^{-(m-i)\mathrm{e}^{-u}}\, \mathrm{d}u.$$

Use $2^{-j\mathrm{e}^{-u}} = 1 - j \log 2\, \mathrm{e}^{-u} + B_j(u)$. The constant term contributes

$$\sum_{i=0}^{m} 2^i = 2^{m+1} - 1.$$

The linear term contributes

$$\begin{aligned} -\log 2 \sum_{i=0}^{m} \frac{2^i (m-i)}{i!} \int_0^\infty u^i \mathrm{e}^{-2u}\, \mathrm{d}u &= -\frac{\log 2}{2} \sum_{i=0}^{m} (m-i) \\ &= -\frac{\log 2}{4} m(m+1). \end{aligned}$$

The remaining term is $\sum_{i=0}^{m} 2^i \eta_i(m-i)$. Subtracting 2^{m+1} and adding $(\log 2) m^2/4$ gives (59). Finally, for $x \geq 0$, $\mathrm{e}^{-x} \geq 1 - x$; applying this with $x = j \log 2\, \mathrm{e}^{-u}$ proves $B_j(u) \geq 0$. □

Lemma 5.4. *For every $m \geq 0$,*

$$|\delta_m| \leq D(m), \qquad D(m) = (\log 2) m \log(m+1) + 1. \tag{60}$$

Proof. For the lower bound, (59) gives $\delta_m \geq -1 - \frac{\log 2}{4} m \geq -D(m)$ for $m \geq 1$, because $\log(m+1) \geq \log 2 > 1/4$; and $\delta_0 = -1 = -D(0)$.

For the upper bound put $\Lambda_m = \sum_{i=0}^{m} 2^i \eta_i(m-i)$. Since $\mathrm{e}^{-x} - 1 + x \leq \min(x, x^2/2)$ for $x \geq 0$, taking $x = (m-i) \log 2\, \mathrm{e}^{-u}$ gives

$$\Lambda_m \leq \int_0^\infty \mathrm{e}^{-u} \sum_{i=0}^{m} \frac{(2u)^i}{i!} \min\left((m-i) \log 2\, \mathrm{e}^{-u}, \frac{(m-i)^2 (\log 2)^2 \mathrm{e}^{-2u}}{2} \right) \mathrm{d}u. \tag{61}$$

Let $v_m = \max\left(0, \log \frac{m+1}{3}\right)$. On $0 \leq u \leq v_m$ use the linear bound together with $\sum_{i=0}^{m} (m-i)(2u)^i/i! \leq m\mathrm{e}^{2u}$, and on $u \geq v_m$ the quadratic bound together with $\sum_{i=0}^{m} (m-i)^2 (2u)^i/i! \leq m^2 \mathrm{e}^{2u}$. Then

$$\Lambda_m \leq m \log 2\, v_m + \frac{m^2 (\log 2)^2}{2} \mathrm{e}^{-v_m}. \tag{62}$$

For $m \geq 2$ we have $v_m = \log \frac{m+1}{3}$ and $\mathrm{e}^{-v_m} = \frac{3}{m+1}$, so (62) and $\frac{3m^2}{2(m+1)} \leq \frac{3}{2}m$ give

$$\begin{aligned}\Lambda_m &\leq m \log 2 \log \frac{m+1}{3} + \frac{3m^2(\log 2)^2}{2(m+1)} \\ &\leq m \log 2 \left(\log(m+1) - \log 3 + \tfrac{3}{2}\log 2\right) \leq m \log 2 \log(m+1),\end{aligned}$$

the last step because $3 > 2^{3/2}$. For $m \leq 1$ we have $v_m = 0$, and (62) reads $\Lambda_m \leq m^2(\log 2)^2/2$, which is at most $m \log 2 \log(m+1)$ since $\frac{m \log 2}{2} \leq \log(m+1)$ there. In both cases

$$\delta_m = -1 - \frac{\log 2}{4} m + \Lambda_m \leq \Lambda_m \leq (\log 2) m \log(m+1) < D(m),$$

which with the lower bound proves (60). □

Lemma 5.5. *If $S^* = \sum_{i \geq 0} 2^i |c_i|$, then, with D as in Lemma 5.4,*

$$|2\Delta(k) - \Delta(k+1)| \leq S^* \big(2D(k) + D(k+1)\big). \tag{63}$$

Proof. Since D is increasing,

$$|\Delta(k)| \leq \sum_{i=0}^{k} 2^i |c_i| D(k-i) \leq S^* D(k).$$

Apply the triangle inequality to $2\Delta(k) - \Delta(k+1)$. □

5.4. **The exponentially small term.** The three terms in (52) are respectively a coefficient tail at $z = 1$, a weighted coefficient tail at $z = 2$, and the integral over $s \geq 2$. The following is the exact numerical input used for them.

Proposition 5.6 (Certified unrestricted input)**.** *The interval computations give*

$$\begin{aligned} d \geq 0.3786949, \qquad S^* \leq 6.9764, \qquad \alpha_+ \leq 15.425661, \\ |U| \leq U_{\max} = 0.08436, \qquad |V| \leq V_{\max} = 1.907462. \end{aligned} \tag{64}$$

For every $k \geq 900$,

$$2^k |T_k| \leq \frac{1.45}{k}, \tag{65}$$

and consequently, as shown at (52) *below,*

$$|E_{\exp}(k)| \leq \frac{1.45}{k} + 10^{-3}. \tag{66}$$

Of the three terms of $E_{\exp}$ only T_k needs a certificate; the other two are bounded analytically in Section 5, and (65) *is recorded here rather than in that derivation so that every certified input appears in one place. The direct certificates establish $f_k < f_{k+1}$ on*

$$6 \leq k \leq 199, \qquad 78 \leq k \leq 899, \qquad 900 \leq k \leq 2439, \tag{67}$$

three separate runs whose union is the single range $6 \leq k \leq 2439$. The quadrature behind these values, and the six error sources entering each enclosure, are described in Appendix A.

The coefficient tails in (66) are obtained only after the pole at $z = 3$ has been removed. Put

$$\widetilde{G}_2(z) = (1 - z/3) G_2(1, z) = \sum_{m \geq 0} w_m z^m.$$

The certified contour bound is $\sup_{|z|=4.2}|\widetilde{G}_2(z)| \le 101.245$. Hence $|w_m| \le 101.245\,4.2^{-m}$, and from $G_2(1,z) = (1-z/3)^{-1}\widetilde{G}_2(z)$ we obtain

$$|c_i| \le 101.245\,3^{-i}\sum_{m=0}^{i}(3/4.2)^m < 354.36\,3^{-i}. \tag{68}$$

By (43) the c_i are the coefficients $h_i(1)$, so Lemma 3.4(i) sharpens (68) to $|c_i| \le \Sigma_0 3^{-i}$, dispensing with the contour bound; the weaker form suffices below. With $C = 354.36$,

$$\begin{aligned} 2^k|L_k| &\le C(2/3)^k, \\ \frac{\log 2}{4}|Q_k| &\le \frac{\log 2}{4}C(2/3)^k\sum_{r\ge 1} r^2(2/3)^r = \frac{30C\log 2}{4}(2/3)^k. \end{aligned}$$

Both terms are less than 10^{-3} in total for $k \ge 900$, and both bounds are geometric in k, so no computation is needed to make them uniform. The third term is the one that needs a certificate, namely (65); substituting it into (52) proves (66), and no asymptotic O-term enters the final criterion. The first two terms are in fact below 10^{-3} from $k \ge 37$ on, so the threshold 900 comes from T_k alone.

6. Assembly of the unrestricted theorem

For $k \ge 900$ define

$$\begin{aligned} R(k) = {} & \frac{\log 2}{4}\big((2k-2)U_{\max} + 2V_{\max} + 4U_{\max}\big) \\ & + S^*\big(2D(k) + D(k+1)\big) + \alpha_+(3k+4) + 3\left(\frac{1.45}{k} + 10^{-3}\right), \end{aligned} \tag{69}$$

and

$$\varphi(k) = a_1(k^2 - 2k - 1) - R(k). \tag{70}$$

Proposition 6.1. *For $k \ge 900$,*

$$2E(k) - E(k+1) \ge -R(k). \tag{71}$$

Consequently $\varphi(k) > 0$ implies $f_k < f_{k+1}$.

Proof. Apply (53) at k and $k+1$. The S_1 contribution is bounded by (56). The signed moment contribution is

$$\frac{\log 2}{4}\big(V + (2-2k)U\big),$$

which is bounded below by (58). The first line of (69) is slightly larger than the resulting quantity, since

$$V_{\max} + (2k-2)U_{\max} \le (2k-2)U_{\max} + 2V_{\max} + 4U_{\max}.$$

The Δ contribution is bounded by (63). Finally, (66) gives

$$|2E_{\exp}(k) - E_{\exp}(k+1)| \le 2\left(\frac{1.45}{k} + 10^{-3}\right) + \left(\frac{1.45}{k+1} + 10^{-3}\right) \le 3\left(\frac{1.45}{k} + 10^{-3}\right).$$

Adding the four lower bounds proves (71). Equation (40) then gives

$$f_{k+1} - f_k \ge 2^{-k-1}\varphi(k),$$

which proves the last assertion. □

Lemma 6.2. *With the certified constants in Proposition 5.6, $\varphi(2440) > 1155$.*

Proof. Substitution in (70) gives the following certified one-sided bounds:

$$\begin{aligned} a_1(2440^2 - 2\cdot 2440 - 1) &> 390371.6,\\ \frac{\log 2}{4}\big(4878U_{\max} + 2V_{\max} + 4U_{\max}\big) &< 72.1,\\ S^*\big(2D(2440) + D(2441)\big) &< 276166.8,\\ \alpha_+(3\cdot 2440 + 4) &< 112977.6,\\ 3\left(\frac{1.45}{2440} + 10^{-3}\right) &< 0.005. \end{aligned}$$

Therefore

$$\varphi(2440) > 390371.6 - 72.1 - 276166.8 - 112977.6 - 0.005 > 1155.$$

The same substitution at $k = 2433$ gives $\varphi(2433) > 135$, so the criterion is already in force there. Direct evaluation puts the sign change just below that, at $\varphi(2432) < 0 < \varphi(2433)$; we do not rely on it, since certifying $\varphi(2432) < 0$ would need a *lower* bound on $R(2432)$, whereas every constant in (64) is one-sided the other way. Nothing below requires the first index: we carry the threshold at 2440, which is one past the end of (67), so that the sweep and the criterion meet without a gap. □

Lemma 6.3. *The function $k \mapsto \varphi(k)$ is strictly increasing for integers $k \geq 2440$.*

Proof. First,

$$a_1\big((k+1)^2 - 2(k+1) - 1 - (k^2 - 2k - 1)\big) = a_1(2k-1).$$

Moreover

$$\begin{aligned} D(m+1) - D(m) &= \log 2\left(\log(m+2) + m\log\left(1 + \frac{1}{m+1}\right)\right)\\ &\leq \log 2\big(\log(m+2) + 1\big). \end{aligned}$$

The last term in (69) decreases with k. Hence

$$\begin{aligned} R(k+1) - R(k) \leq{}& \frac{\log 2}{2}U_{\max} + 3\alpha_+\\ &+ S^*\log 2\big(2\log(k+2) + \log(k+3) + 3\big). \end{aligned} \tag{72}$$

Define, for real $x \geq 2440$,

$$\begin{aligned} M(x) = a_1(2x-1) &- \frac{\log 2}{2}U_{\max} - 3\alpha_+\\ &- S^*\log 2\big(2\log(x+2) + \log(x+3) + 3\big). \end{aligned}$$

Then

$$M'(x) = 2a_1 - S^*\log 2\left(\frac{2}{x+2} + \frac{1}{x+3}\right) > 0,$$

because the certified lower bound for d gives $2a_1 > 0.1312$, whereas the subtracted term is < 0.006 for $x \geq 2440$. Direct substitution gives $M(2440) > 146$. Thus $M(k) > 0$ for every $k \geq 2440$. Together with (72), this yields $\varphi(k+1) - \varphi(k) > 0$. □

Proof of Theorem 1.1. By Lemmas 6.2 and 6.3, $\varphi(k) > 0$ for every $k \geq 2440$. Proposition 6.1 therefore gives $f_k < f_{k+1}$ on that range. The certified ranges in (67) cover every integer from 6 through 2439. Together the analytic and certified ranges cover all $k \geq 6$. □

7. A general principle

Conjecture 7.1. Let $\mathcal{P}$ be a set of primes with least element q and thin complement,

$$\sum_{p\notin\mathcal{P}} \frac{\log p}{p} < \infty, \tag{73}$$

and let $f_k^{\mathcal{P}}$ be the sum of $1/(n\log n)$ over integers n with $\Omega(n)=k$ and all prime factors in $\mathcal{P}$. The sequence $f_k^{\mathcal{P}}$ is eventually increasing if and only if $q=2$.

Some hypothesis of this kind is needed. For $\mathcal{P}=\{2\}$ one has $f_k^{\mathcal{P}} = 1/(2^k k\log 2)$, and for $\mathcal{P}=\{2,3\}$ the sequence again decreases to 0; in both cases $q=2$ and the conclusion fails. What goes wrong is that a sparse $\mathcal{P}$ leaves $\sum_{p\in\mathcal{P}} p^{-s}$ bounded as $s\to 1^+$, so the generating function has no $\zeta(s)^z$ factor and $f_k^{\mathcal{P}}$ decays geometrically to 0 whatever q is.

Under (73) that cannot happen, and the singularity structure is the one the two theorems above exploit:

$$\prod_{p\in\mathcal{P}}(1-zp^{-s})^{-1} = \prod_{p}(1-zp^{-s})^{-1}\prod_{p\notin\mathcal{P}}(1-zp^{-s}),$$

where the second product converges locally uniformly by (73) and is therefore entire in z. Its zeros cancel exactly the poles at $z=p^s$ with $p\notin\mathcal{P}$, leaving the first z-singularity at $z=q^s$. At $s=1$ the point $z=2$ lies on that singular circle exactly when $q=2$. If $q\geq 3$ the coefficient extraction at $z=2$ is interior and gives a constant multiple of 2^{-k}, as in Section 3; if $q=2$ it takes place at the boundary and may produce the factor k^2.

Three points remain to turn this mechanism into a theorem for arbitrary $\mathcal{P}$ satisfying (73). First, the regularized product must provide enough uniform control to identify the limiting constant and the coefficient at $z=2$. Second, in the boundary case one must rule out cancellation of the k^2 coefficient. Third, in the interior case one must show that the constant multiplying 2^{-k} is positive, which is what makes the sequence decrease rather than increase; for $\mathcal{P}=\{p : p\geq 3\}$ this is the inequality $\beta>0$ of (5). The present paper proves the two extremal cases $\mathcal{P}=\{\text{all primes}\}$ and $\mathcal{P}=\{p : p\geq 3\}$.

8. Certification and formalization

What the formalization establishes is an *implication*. The certified quantities enter as fields of five hypothesis structures — `NumericalFacts`, `NumericalFactsFull`, `OddNumericalFacts`, `OddCoeffFacts`, `OddDomFacts` — and no instance of any of them is constructed. Lean therefore checks that those hypotheses imply the theorems, and nothing in the kernel checks that the hypotheses are true, or even that they are mutually consistent: an unsatisfiable field would make the theorems assuming it vacuous rather than false, and would not be detected. That the imported intervals agree with what Arb computed is verified outside Lean, by the script `verify.py` against the certificate. The reader should take the Lean development as discharging the reductions below and nothing else; the trusted base is the structure fields, and it is listed in Table 1.

With that said, the division is as follows. Formalized in Lean 4 without `sorry` are Lemmas 2.1 and 2.3, the coefficient identities (13) and (53), the assembly of the ranges in both theorems, and the lemmas named in the rest of this section.

Two items a reader would expect to find among them are not there, and we state the position precisely. First, Proposition 2.2: a radius-specific consequence of it is proved, on the ball $|z|<3$ for $s\geq 1$, but its operative content (10) — that $\mathrm{d}/\mathrm{d}s$ commutes with $[z^j]$ — is not, and the s-differentiability of the coefficient functions is itself a hypothesis, as field `g_deriv` of `NumericalFactsFull` in the full case and as `deriv_exists`, `deriv2_exists` of

`OddCoeffFacts` in the odd case. No interval computation can establish a differentiability claim, and none is attempted. Second, Proposition 3.5: its statement is imported as field `remainder` of `OddNumericalFacts` rather than derived from Proposition 3.3; the derivation is the proof given in Section 3 above, which has not been formalized. The Lean development contains the intended route to it, but that route still rests on assumed coefficient and dominating-sequence data of its own, so it does not discharge the hypothesis.

Likewise the source estimates of Section 5 are assumed, not proved, by the theorem named below: `NumericalFacts` carries $|2E(k) - E(k+1)| \le R(k)$ as the single field `source_bound`, which is the aggregate of that whole section. Only the second formal version discussed below derives it, from the four source terms separately.

The numerical propositions are Proposition 3.3 and Proposition 5.6. Their statements are finite systems of interval inequalities, established by Arb/FLINT through its Python bindings. The Lean formalization imports the resulting intervals as hypotheses — together with Proposition 3.5 and the differentiability noted above, neither of which is an interval inequality — and proves that those hypotheses imply Theorem 1.1 and both assertions of Theorem 1.2; the three formal statements are `erdos_sums_increase_from_six`, `fOdd_decreasing` and `fOdd_tendsto_half`. The differentiability field belongs to `NumericalFactsFull` and so is a hypothesis of the second, primitive version of the full theorem rather than of `erdos_sums_increase_from_six`, which does not mention it. The limit $g_k \to \frac{1}{2}$ was formerly left unformalized, the monotonicity argument making no use of it; it is now proved there as well, and from the same hypothesis, since (4) already bounds $|g_k - \frac{1}{2} - \beta 2^{-k}|$ by a multiple of $(k+3)^3 3^{-k}$, and both that and 2^{-k} vanish.

For Theorem 1.1 the hypotheses are not only the interval bounds of Proposition 5.6. Proposition 6.1 enters as a hypothesis in one of the two formal versions, and is proved in the other, from the primitive certified quantities rather than from the aggregate bound (71). (The bound (66) is not what the primitive route replaces; it remains a hypothesis, `NumericalFactsFull.Eexp_bound`, in both versions.) What the formalization proves from these is the passage to $f_k < f_{k+1}$ for $k \ge 2440$, together with the assembly of the four ranges.

Lemmas 6.2 and 6.3 were formerly hypotheses of that development as well. They are now proved in it, as `phiBound_pos_at_2440` and `phiBound_mono_of_bounds`, from upper bounds on the parameters alone: φ is an explicit expression in k, so neither lemma is a statement about primes, contours or limits, and what the formalization had been assuming was arithmetic. Proving them needed one quantity the certified hypotheses did not carry, the bound $|U| \le U_{\max}$ of (64), which is now among them. The sharpness required is not uniform: the base case turns on $\log 2441$ to a tenth of a percent, where $\log 2441 \le 12 \log 2$ is already too weak, while the monotonicity step needs only $\log 2441 \le 12 \log 2$ itself.

For the odd theorem the lower bound on γ was likewise a hypothesis and is now proved, as `gamma_lower_bound`, from $\mathrm{harmonic}(n) - \log(n+1) < \gamma$ at $n = 8191$, chosen so that $n+1 = 2^{13}$ and the logarithm is $13 \log 2$ exactly.

The constants d of (3) and ρ were hypotheses of both developments and are now proved in them, as `dConst_lower_bound`, `dConst_lower_bound_sharp` and `rhoConst_upper_bound`. What makes this possible is that the Euler factors are rational and exceed 1,

$$\left(1 - \tfrac{2}{p}\right)^{-1} \left(1 - \tfrac{1}{p}\right)^{2} = \frac{(p-1)^2}{p(p-2)} = 1 + \frac{1}{p(p-2)},$$
$$\left(1 - \tfrac{3}{p}\right)^{-1} \left(1 - \tfrac{1}{p}\right)^{3} = \frac{(p-1)^3}{p^2(p-3)} = 1 + \frac{3p-1}{p^2(p-3)},$$

the second because $(p-1)^3-p^2(p-3)=3p-1$. The constant regularized by the second display is

$$\rho=-\frac{1}{9}\prod_{p\geq 5}\Big(1-\frac{3}{p}\Big)^{-1}\Big(1-\frac{1}{p}\Big)^{3}, \tag{74}$$

which is recorded here only so that every constant proved in the formalization has a counterpart in the text; it is used nowhere in the proofs above. A product of factors each at least 1 is bounded below by the product over any finite subset of them, with no tail estimate entering, so a one-sided bound on such a constant is a finite computation in the kernel even where the two-sided enclosure is not. Only one-sided bounds are used: d enters through $a_1=d\log 2/4$ and through $\beta=d(\log 2-S+\gamma)$, in both cases from below, and ρ is not used at all. The bounds proved are weaker than those of Proposition 5.6, which are unchanged. The 94 primes $p\leq 499$ give $d\geq 0.3785913$, which suffices for $a_1\geq 0.0656$; the 668 primes $p\leq 4999$ give $d\geq 0.3786823$, which is what $\beta\geq 0.1556$ requires; and the 93 primes $5\leq p\leq 499$ give $\rho\leq -0.1747885$. In each case the requirement is computed against the bound for $\log 2$ available in the formalization rather than against $\log 2$ itself, the two differing by enough to change how many primes are needed.

The same device does not reach S. There the bound required is an upper one on a sum of positive terms, so the tail cannot be discarded, and the summands carry $\log p$ rather than being rational. The budget is narrow: with d at the bound just proved, $\beta\geq 0.1556$ permits $S\leq 0.8593986$, while the partial sum over $p<2\cdot 10^6$ already reaches 0.8593917, so the admissible tail is below $7\cdot 10^{-6}$ and primes beyond about $1.5\cdot 10^5$ are needed. Accordingly S remains a certified hypothesis, as do the sweeps and the Cauchy estimates of Propositions 3.3 and 5.6.

Two constants are stated above more conservatively than the computation requires, so that the text and the formalization carry the same values: the threshold 2440 of Lemma 6.2 in place of 2433, and (68) in place of the sharper bound that Lemma 3.4(i) supplies. Neither changes the ranges covered. Two further discrepancies should be recorded rather than left for the reader to find. The certified odd sweep is stated above for $1\leq k\leq 75$, whereas the field `OddNumericalFacts.sweep_low` assumes only $1\leq k\leq 58$; since the analytic branch begins at $k=59$ that is all the assembly needs, and assuming less is the safe direction, but the two ranges are not the same. And the field `NumericalFacts.Gamma_bound`, carrying $\Gamma\leq 104.52$, has no counterpart above and no consumer in the development either; it is retained only so that the certified package and the text can be compared field by field.

For the odd theorem, the coefficient bounds in (25) are obtained by cancelling the pole at $z=3$ before applying coefficient recursion or Cauchy bounds. The range $0<u\leq 3$ is evaluated directly, while the range $u\geq 3$ uses the Taylor model made explicit in Proposition 3.1.

The constant C_E is certified on the whole of $(1,s(3)]$, and in fact on its closure: the four quantities entering (24) are ratios in which the pole of ζ at $s=1$ cancels, so each extends continuously to $s=1$, and the Taylor model below is centred at $1+E/2$ with radius $E/2$ and therefore covers the closed interval $[1,s(3)]$ on which Σ_2 takes its own supremum. Interval evaluation on boxes of s loses that cancellation, since each occurrence of $s-1$ is widened independently, and the enclosures diverge however finely $(1,s(3)]$ is partitioned. Expanding ζ as a power series in $\sigma=s-s_0$ about $s_0=1+E/2$, where $E=s(3)-1$, makes the cancellation symbolic. The resulting series are evaluated on $|\sigma|\leq E/2$, which is the whole range; their coefficients fall off geometrically, and truncating after nine terms contributes less than 10^{-20}.

The constant C_N requires $\Pi(t)=\sum_{p\geq 3}p^{-t}$ at $t=s,2s,\dots,200s$ for $s=s(3)$. At $s\approx 1.05$ these converge too slowly to truncate: the Rosser–Schoenfeld bound applied beyond 10^8 still overestimates the tail twofold, and that excess compounds through the coefficient recursion.

TABLE 1. What the Lean development proves and what it assumes. The assumed rows are the trusted base: no instance of any structure is constructed, so these are not checked in the kernel.

Statement	Status	Lean name
Theorem 1.1	proved	`erdos_sums_increase_from_six`
Theorem 1.2, monotonicity	proved	`fOdd_decreasing`
Theorem 1.2, limit	proved	`fOdd_tendsto_half`
Lemma 6.2	proved	`phiBound_pos_at_2440`
Lemma 6.3	proved	`phiBound_mono_of_bounds`
γ from below	proved	`gamma_lower_bound`
d from below	proved	`dConst_lower_bound`, `_sharp`
ρ from above	proved	`rhoConst_upper_bound`
Proposition 3.5	assumed	`OddNumericalFacts.remainder`
Section 5 aggregate	assumed	`NumericalFacts.source_bound`
(66)	assumed	`NumericalFactsFull.Eexp_bound`
g_i differentiable on $(1,2)$	assumed	`NumericalFactsFull.g_deriv`
∂_s, ∂_s^2 of coefficients exist	assumed	`OddCoeffFacts.deriv_exists`, `deriv2_exists`
S from above	assumed	`NumericalFacts.Sstar_bound`
Sweeps	assumed	`OddNumericalFacts.sweep_low` and companions
Σ_i, C_E, C_N, N_U, α_+, U, V	assumed	fields of the five structures

Instead Π is evaluated by the Möbius series $\Pi(t) = \sum_{n\ge1} \mu(n) n^{-1} \log\zeta(nt) - 2^{-t}$, which requires no prime enumeration. The recursion itself is run at 600 bits, since cancellation costs about one digit every 2.3 steps over the 200 terms used; at 1200 bits the head agrees to every printed digit.

For the unrestricted theorem, the s-uniform coefficient bounds defining α_+ are obtained by carrying s as an interval over a partition of $[1,2]$. The signed moments U and V are evaluated before absolute values are taken, for the reason given at (57).

Availability and versions. Since the contribution is in part a certification, the artefacts and the versions they were produced with are part of the claim. The Lean development, the Arb scripts, their logs and the certificate are at `https://github.com/jaideepsaipadhi/erdos-sums-almost-primes`. The formalization is built with Lean 4, toolchain `leanprover/lean4:v4.33.0-rc1`, against Mathlib at revision `79d0395a1825`; the permitted axioms are exactly `propext`, `Classical.choice` and `Quot.sound`, which is checked by `#print axioms` on each named theorem. The numerics use python-flint 0.9.0 against FLINT 3.6.0. The certificate records the parameters of every run together with a SHA-256 hash of the numerical payload, and `verify.py` checks the imported intervals against it.

APPENDIX A. CERTIFIED QUADRATURE

Since $1/(n\log n) = \int_1^\infty n^{-s}\,\mathrm{d}s$, we have $f_k = \int_1^\infty P_k(s)\,\mathrm{d}s$, and the substitution $s = 1 + \mathrm{e}^{-u}$ turns this into

$$f_k = \int_{-\infty}^{\infty} F(u)\,\mathrm{d}u, \qquad F(u) = P_k(1+\mathrm{e}^{-u})\mathrm{e}^{-u}, \tag{75}$$

the integrand being analytic on the strip $|\operatorname{Im} u| < \pi/2$. The integral is split into a truncated range $[u_-, u_+]$ with $u_- = -5.10$ and $u_+ = 3.5k_{\max}$, taken once for the largest k of the run and so uniform in k, and the two tails beyond it. The truncated range is divided into panels of width $H = 0.8$, each mapped affinely to $[-1, 1]$ and integrated by Gauss–Legendre quadrature with $N = 52$ nodes at 1400 bits of precision; the number of panels therefore grows linearly with $k_{\max}$.

Four of the six error sources in the certificate are arithmetic: ball arithmetic throughout, the explicit truncation balls in the Möbius series for the prime zeta function, the discretization bound derived below, and interval-Newton enclosures for the Gauss nodes and weights. The remaining two are the tails of (75), which are bounded as follows and are not negligible by inspection.

For the left tail, $u \le u_-$ corresponds to $s \ge s_- = 1 + \mathrm{e}^{5.10} = 165.02$. There $P_1(s) = \sum_p p^{-s} \le 2.78 \cdot 2^{-s}$, since $P_1(s)2^s = 1 + \sum_{p \ge 3} (2/p)^s$ decreases in s and is already below 2.78 at $s = 2$; and $P_k(s) \le P_1(s)^k$, because expanding $P_1(s)^k$ counts every n with $\Omega(n) = k$ at least once. Hence $P_k(s) \le 2.78^k 2^{-ks}$, and

$$\int_{-\infty}^{u_-} |F| \, \mathrm{d}u = \int_{s_-}^{\infty} P_k(s) \, \mathrm{d}s \le \frac{2.78^k 2^{-ks_-}}{k \log 2}. \tag{76}$$

For the right tail, $P_k(s) \le (\log \zeta(s))^k / k!$ for $s > 1$ [2, Lemma 5.6], so with $u = \log \zeta(s)$ the integral is dominated by $\int_{u_+}^{\infty} u^k \mathrm{e}^{-u} \, \mathrm{d}u / k!$; the resulting incomplete-Gamma integral obeys $\Gamma(k+1, u_+)/k! \le 2u_+^k \mathrm{e}^{-u_+}/k!$ provided $u_+ \ge 2k + 2$, so

$$\int_{u_+}^{\infty} |F| \, \mathrm{d}u \le \frac{2u_+^k \mathrm{e}^{-u_+}}{k!}. \tag{77}$$

The hypothesis $u_+ \ge 2k + 2$ holds throughout, since $u_+ = 3.5k_{\max}$ and $k \le k_{\max}$. Both bounds are added to every reported enclosure. They are largest at $k = k_{\max}$, and even there they are far below the differences $f_{k+1} - f_k \asymp a_1 k^2 2^{-k}$ that the sweep has to resolve: for $k_{\max} = 199$ the right-hand side of (77) is below 10^{-100} while the smallest difference in range exceeds 10^{-57}.

For f analytic inside the Bernstein ellipse E_ρ and bounded there by M, the N-node Gauss rule satisfies

$$\left| \int_{-1}^{1} f(x) \, \mathrm{d}x - \widehat{I}_N \right| \le \frac{64}{15} \frac{M \rho^{-2N}}{1 - \rho^{-2}} \tag{78}$$

[5, Thm. 4.5], where the rule is indexed so that I_n uses $n + 1$ nodes and the stated exponent $\rho_B^{-(2n+2)}$ is the ρ_B^{-2N} above; the bound appears as [6, Thm. 19.3]. Here ρ_B is the Bernstein-ellipse parameter, unrelated to the arithmetic constant ρ of Section 8. The implementation evaluates the right side of (78) with the interval-scaling factor $H/2$ and a further factor 10 of margin. The ellipse is taken with semi-minor axis $\delta = 1.45 < \pi/2$ in the u variable, giving $\rho_B \approx 7.385$, and M is bounded using $|P_k(s)| \le P_k(\operatorname{Re} s)$ together with the monotonicity of P_k on the reals.

The per-panel bound is not monotone in the panel's position, and it would be wrong to locate its maximum by inspection: on the right-hand side $F(u) \approx u^k \mathrm{e}^{-u}/k!$ by (75), which increases until $u \approx k$ and only then decays, so M rises across the bulk of the range before falling. What is used is therefore the maximum over all panels and all $k \le k_{\max}$, which the certificate records; it is 1.1×10^{-87}, and so is negligible beside the widths actually reported. It is also, at $k_{\max} = 199$, some twenty orders of magnitude larger than either tail (76)–(77), so the discretization bound and not the truncation is what limits the accuracy of the f_k.

The pinned coefficient sweeps use primes up to 10^5 and 300 bits of precision, with 1600 boxes in s for α_+ and 800 boxes for the second-derivative sweep. The certificate records these parameters together with a SHA-256 hash of the numerical payload.

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Statement on the use of AI

The analytic reductions, the interval-arithmetic certificates, the Lean development and the text of this paper were all produced with extensive use of an AI assistant (Anthropic's Claude). I chose the problem, directed the work throughout, and take full responsibility for every claim made here, including those I reached with that assistance.

In more detail, the assistant was used to derive and check the reductions of Sections 2–6; to write and run the Arb/FLINT scripts that produce the certified constants of Propositions 3.3 and 5.6; to write the Lean development described in Section 8; and to draft and revise this manuscript. It was also used adversarially, to look for errors in work already believed finished. That is how the defect in the near-range step of Proposition 3.5 was found: the step bounded a sum of absolute values by a signed coefficient identity, which is invalid because the h_j alternate in sign. It is the reason that proposition is stated on $k \geq 59$.

Two limits on what the foregoing should be taken to mean. The numerical claims do not rest on the assistant's reliability: they are certified by interval arithmetic and are reproducible from the scripts and logs in the repository cited in Section 8. The Lean development, by contrast, establishes an implication from assumed hypotheses rather than a closed proof, as Section 8 and Table 1 set out in detail.

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